# MCAR Missingness – Some Options & Considerations

Two Fundamental Paradigms

**1. Univariate Imputation**

- Uses **only the historical information of the target variable e.g.**
  - mean / interpolation, Kalman smoothing, Brownian bridge, ARIMA-based methods
- Primary objective: Preserve **temporal continuity and dynamics**
- Simple & interpretable
- Robust when cross-sectional information is weak or unstable

**Limitations:**

- Ignores contemporaneous cross-variable relationships
- Can understate uncertainty during long gaps

**2. Multivariate Imputation**

- Uses **other variables observed at the same time points**
  - KNN, MICE, Random Forests, XGBoost, neural imputers (SAITS, MIDA)
- Primary objective:
- Preserve **cross-sectional and structural relationships**
- Exploits shared information across related series
- Often superior for large gaps when peers are informative

**Limitations:**

- Sensitive to feature quality and missingness structure
- Risk of “relationship leakage” if used improperly in modelling pipelines

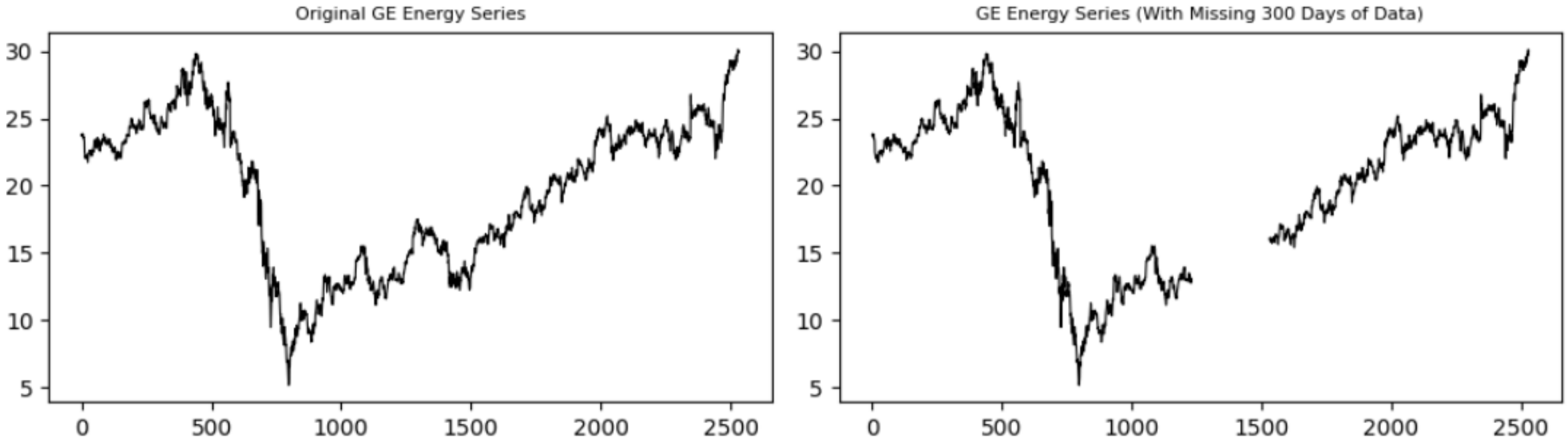
**Key to consider: Temporal dynamics, Volatility structure, Visual plausibility, Edge gaps vs. interior gaps….**

# Original Data

| | 0 | 1 | 2 |
|---|---|---|---|
| **ALJ** | 14.818392 | 14.825764 | 15.142775 |
| **APC** | 45.257976 | 43.982260 | 45.543516 |
| **ATW** | 20.461035 | 20.257732 | 20.686121 |
| **BBG** | 40.500000 | 39.010000 | 38.930000 |
| **BC** | 36.190061 | 35.994775 | 36.367594 |
| **...** | ... | ... | ... |
| **WLB** | 23.840000 | 24.000000 | 24.490000 |
| **WLL** | 21.110000 | 20.750000 | 21.390000 |
| **WTI** | 23.047323 | 23.364905 | 23.720293 |
| **WWD** | 13.637162 | 13.664845 | 14.090862 |
| **XOM** | 43.558401 | 43.342729 | 44.197981 |

65 rows × 3 columns

# Original Data – Artificially Created Data Gap := 300 days



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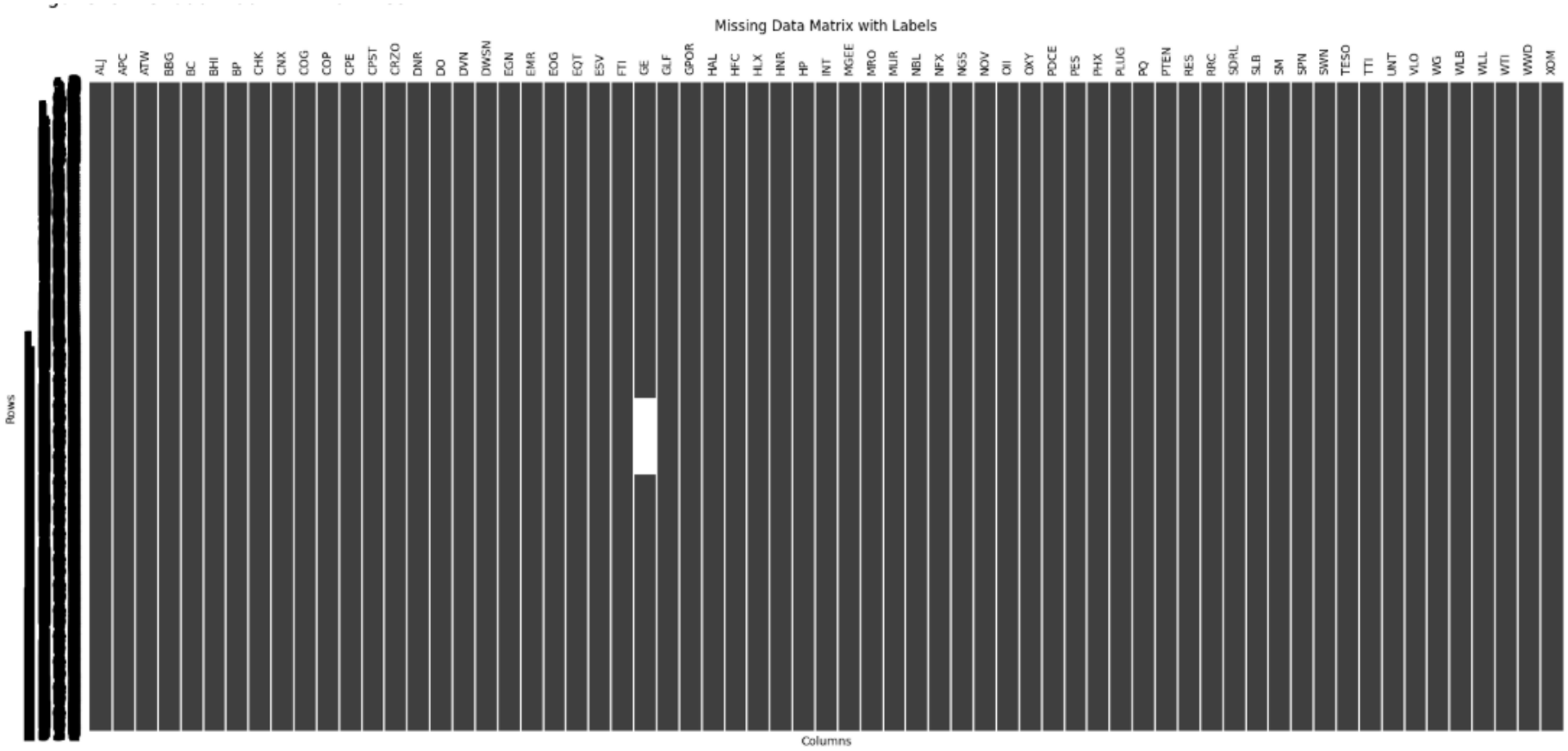
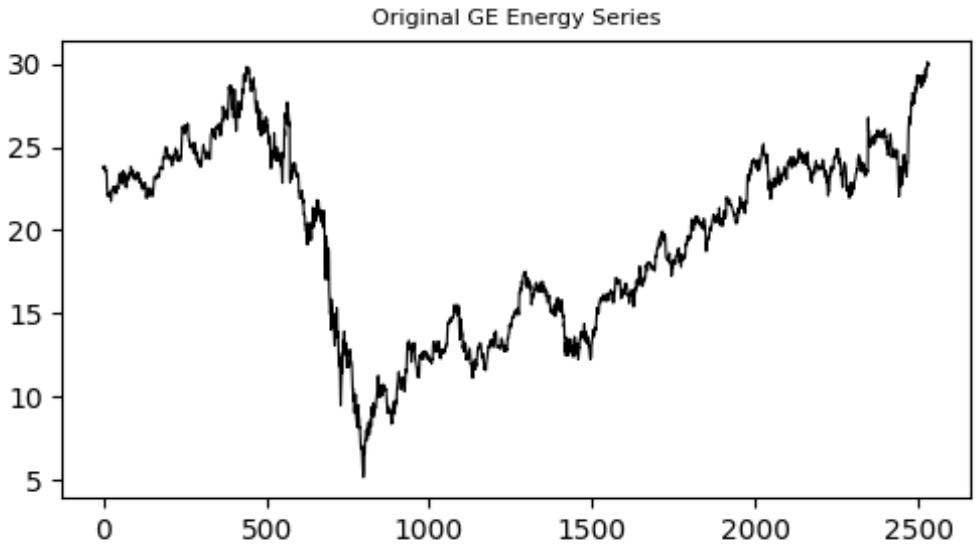
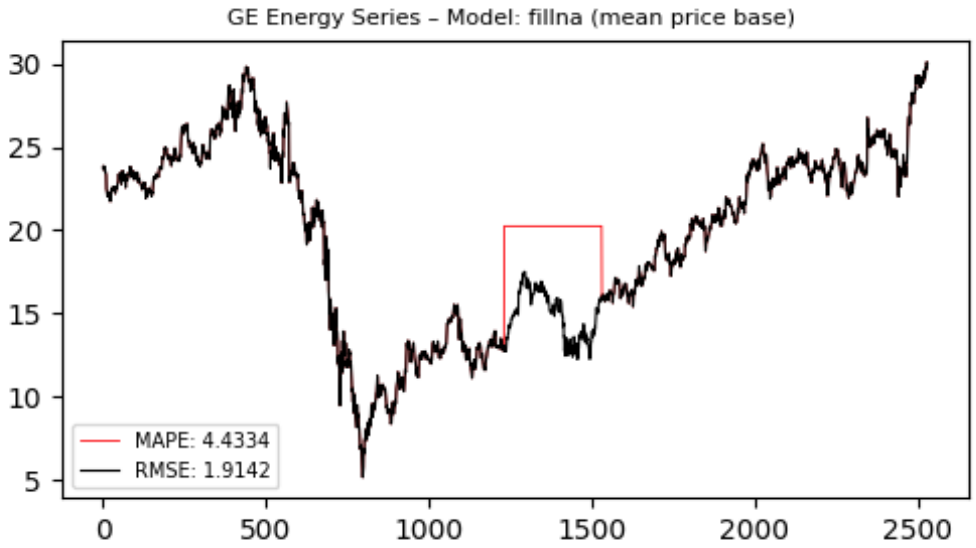


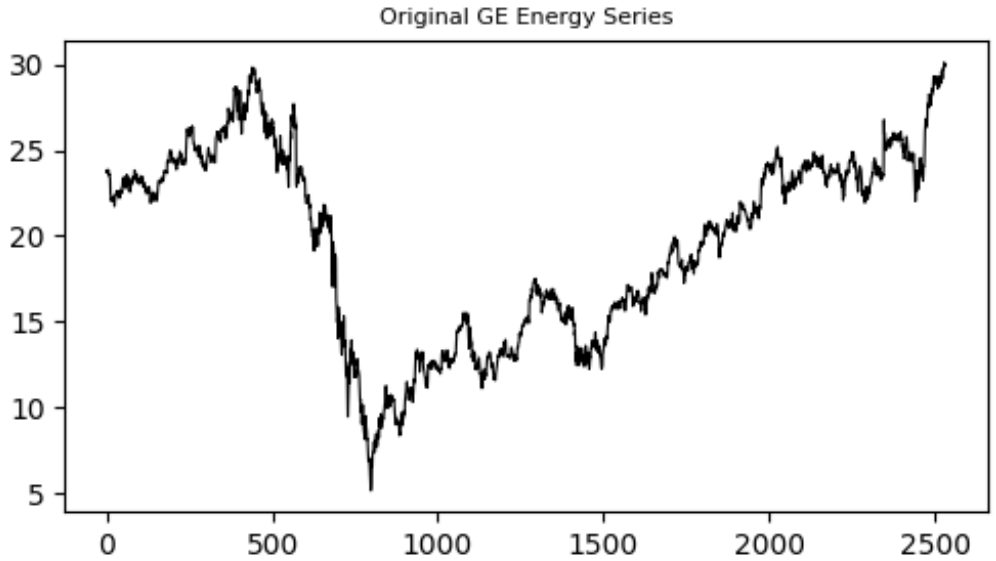
# MCAR Performance Metrics - Notes

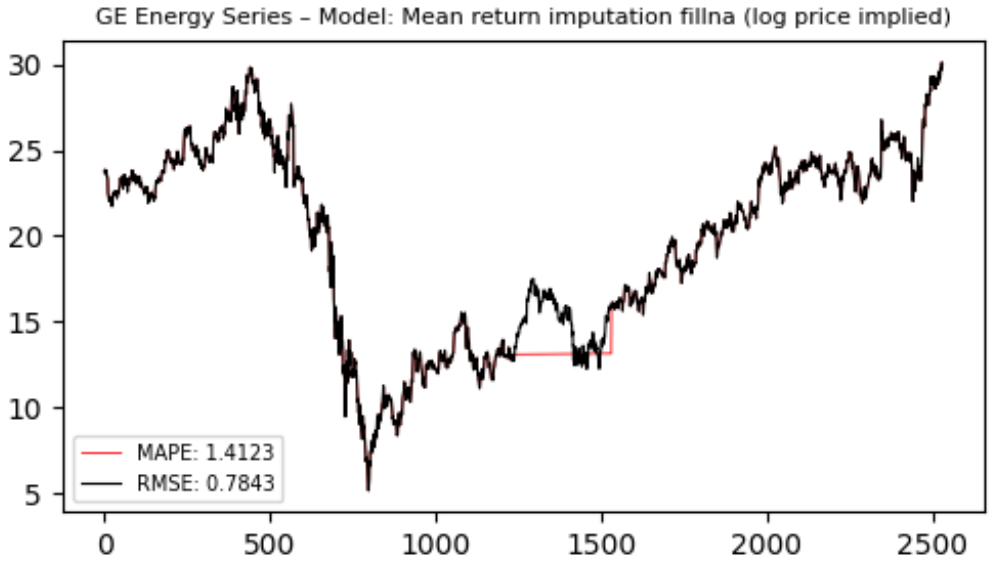
1. Total series length : 2532 rows
2. Gap : 300 days
3. Non-gap share of the data : 2232/2532 $\approx$ 88.1%
4. Gap share = 300/2532 $\approx 11.9\%$

# Model – fillna (data) - Price & Log Return Data

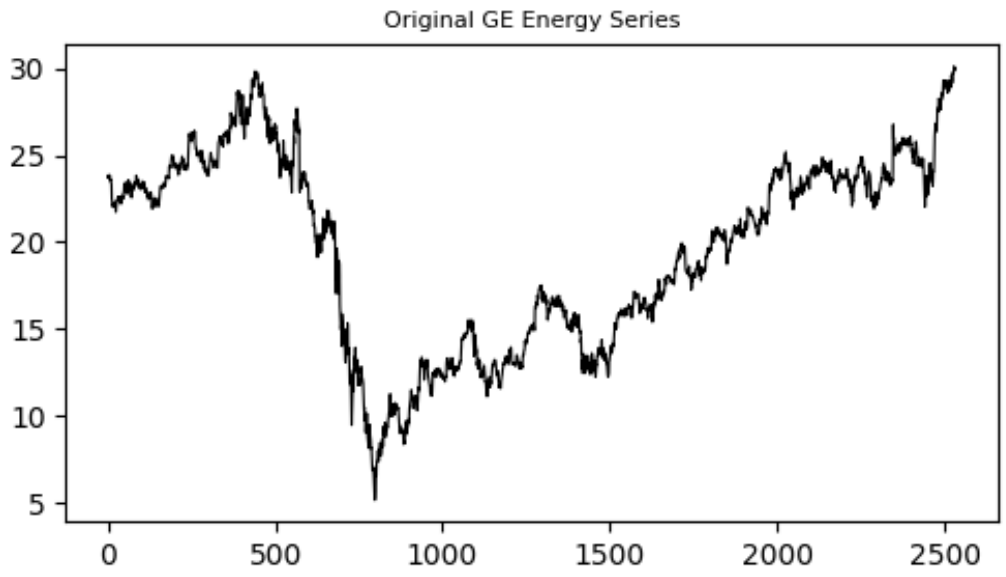


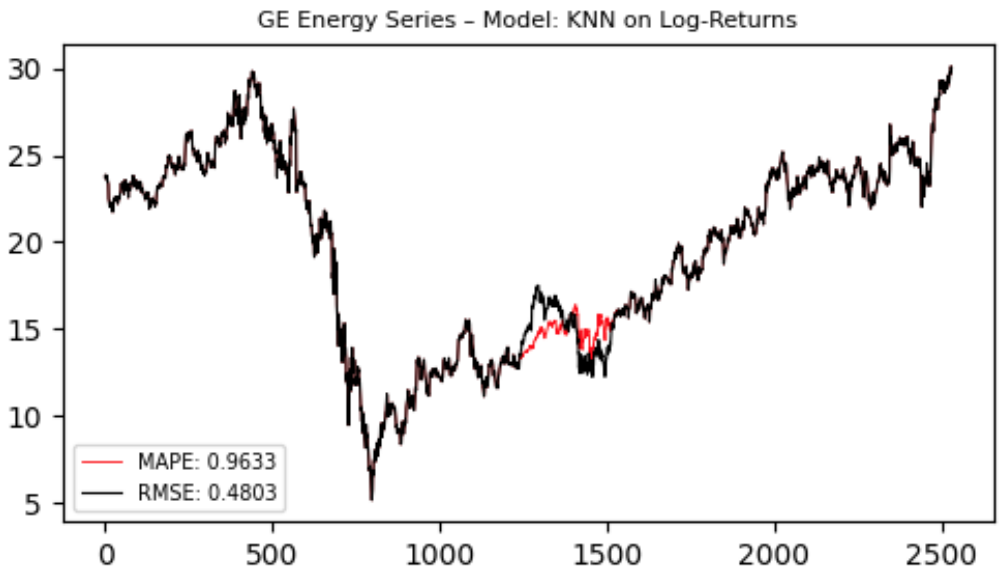



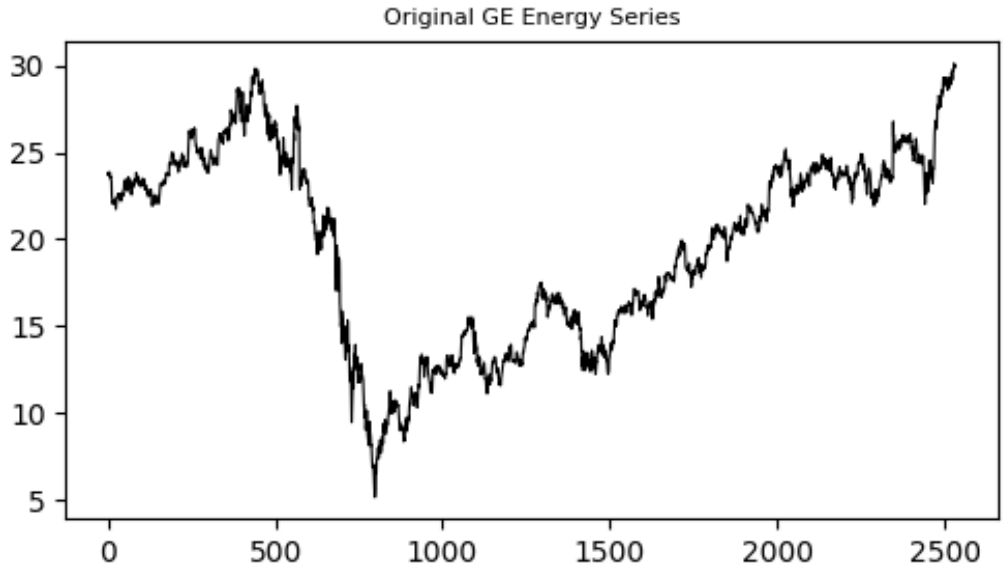



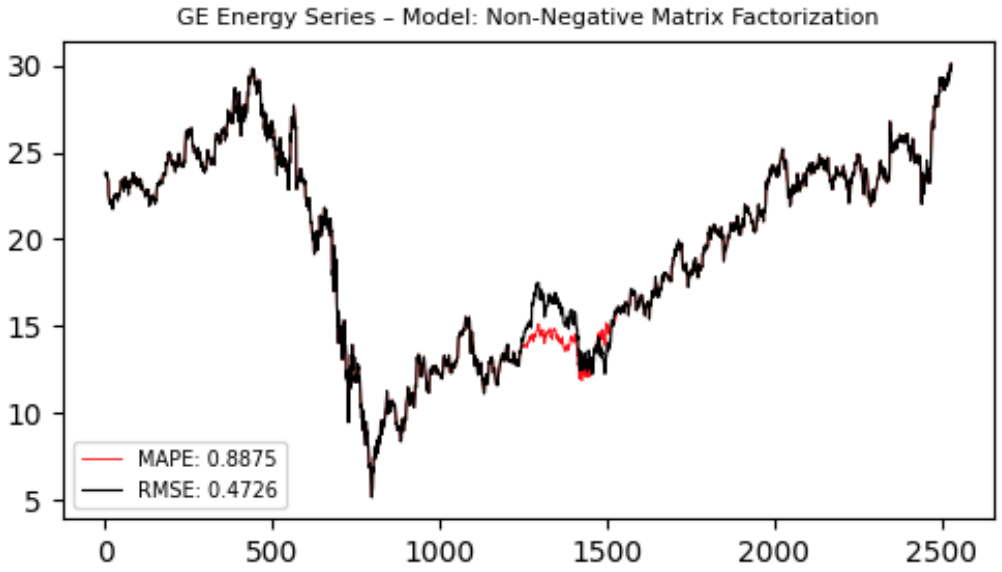



# Model – KNN & Non-Negative Matrix Factorization (NMF)

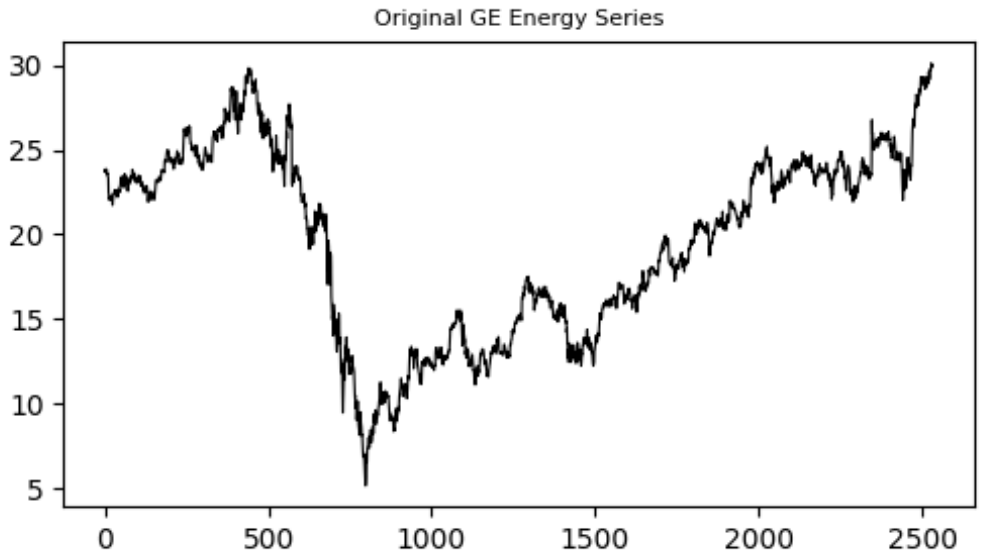


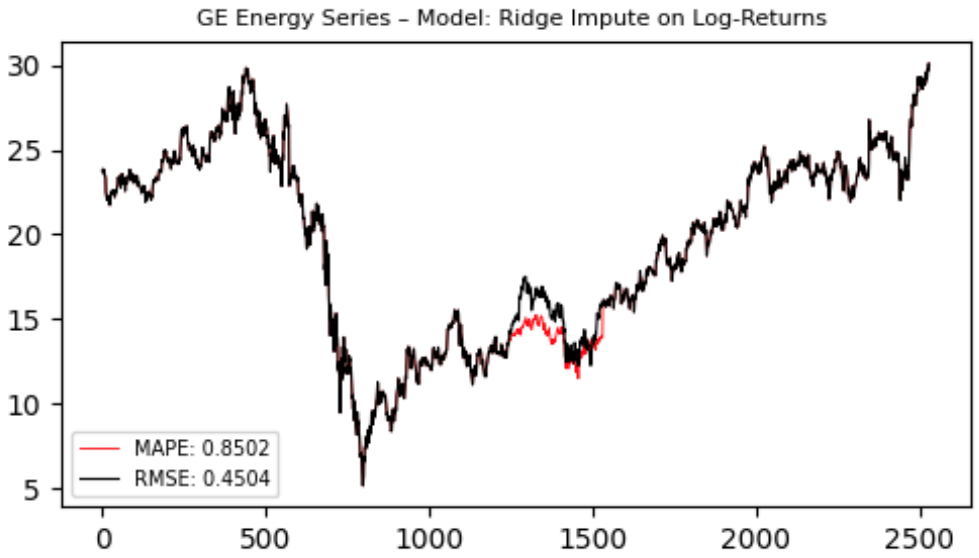



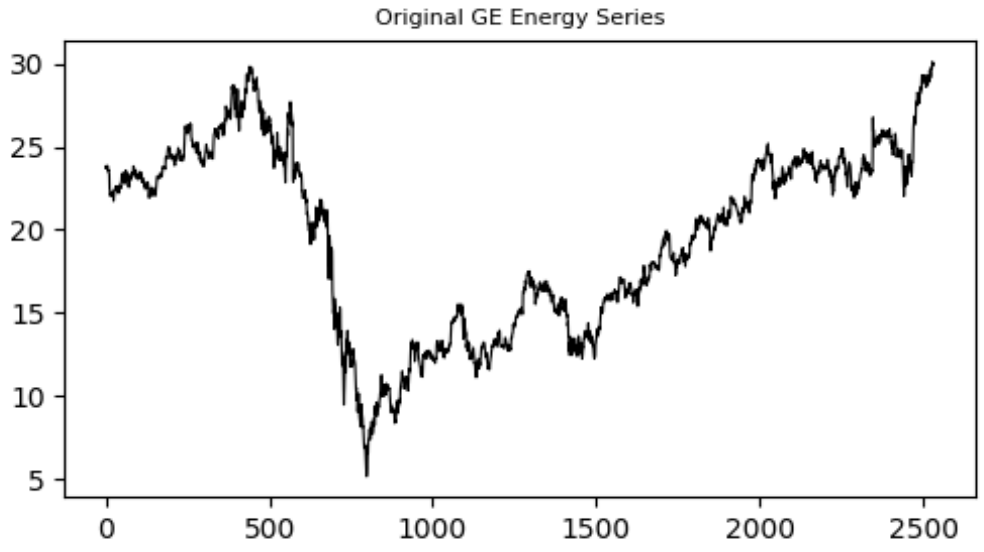



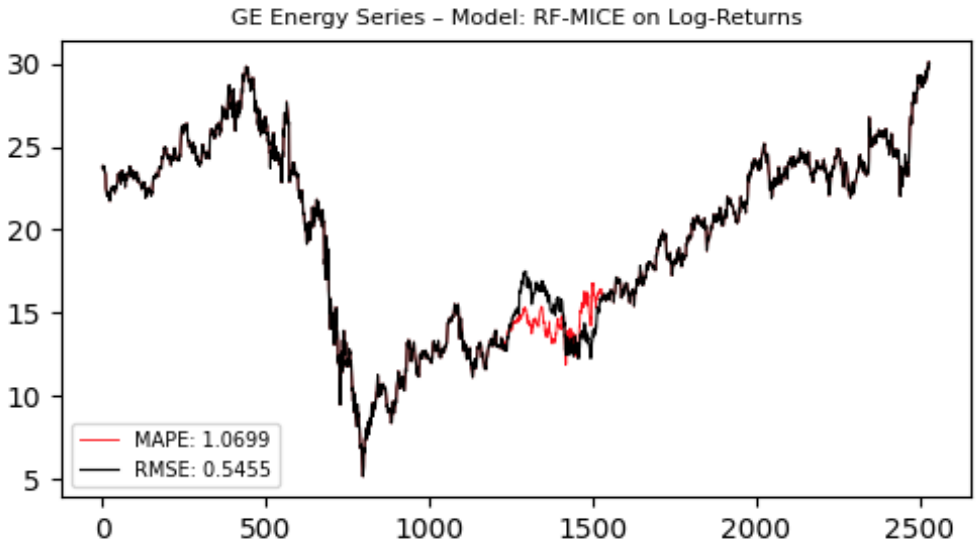



# Model – Iterative Regression & Random Forest Approach

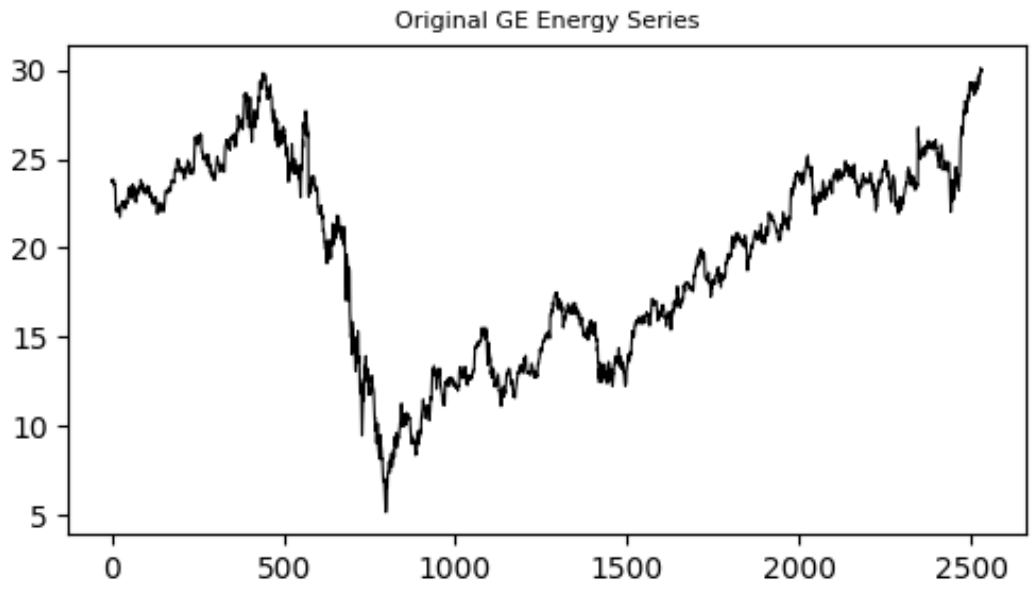


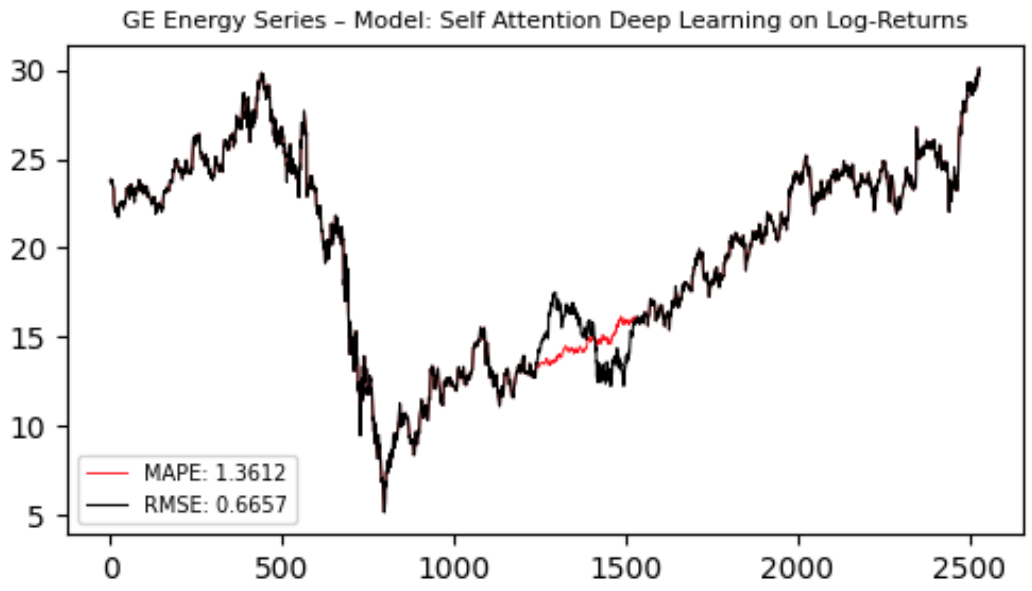



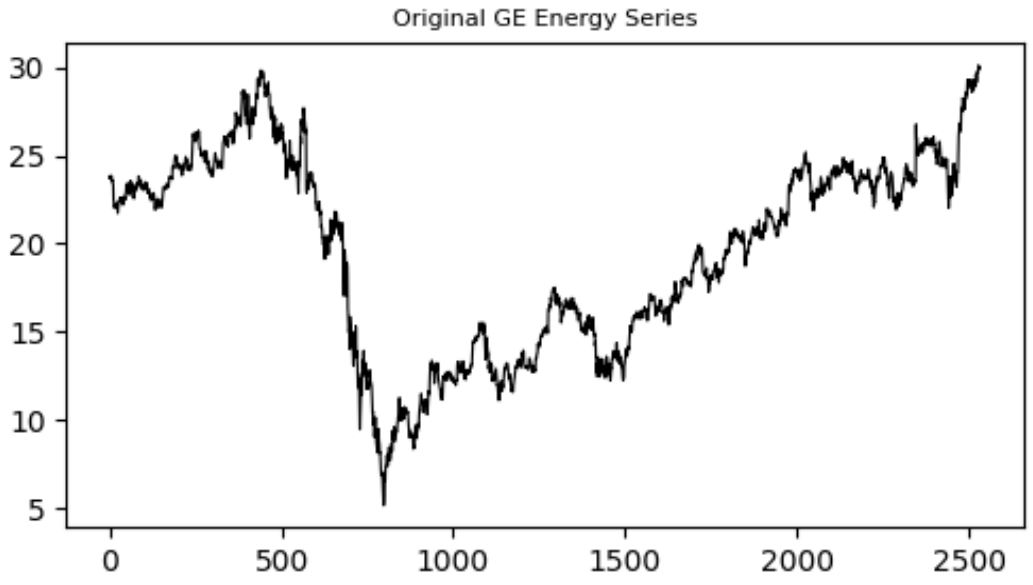



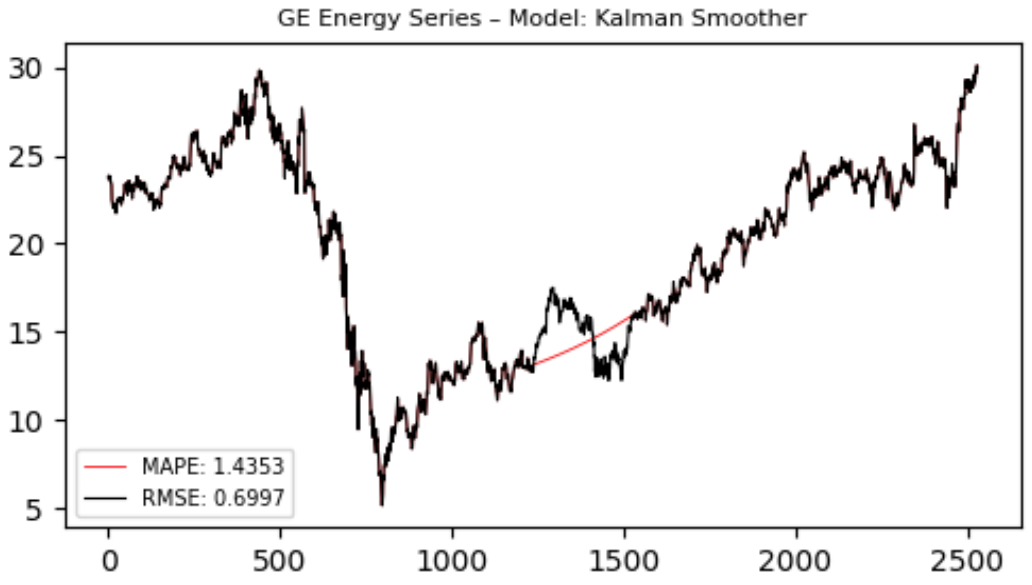



# Model – Self Attention/Deep Learning & Kalman Smoother

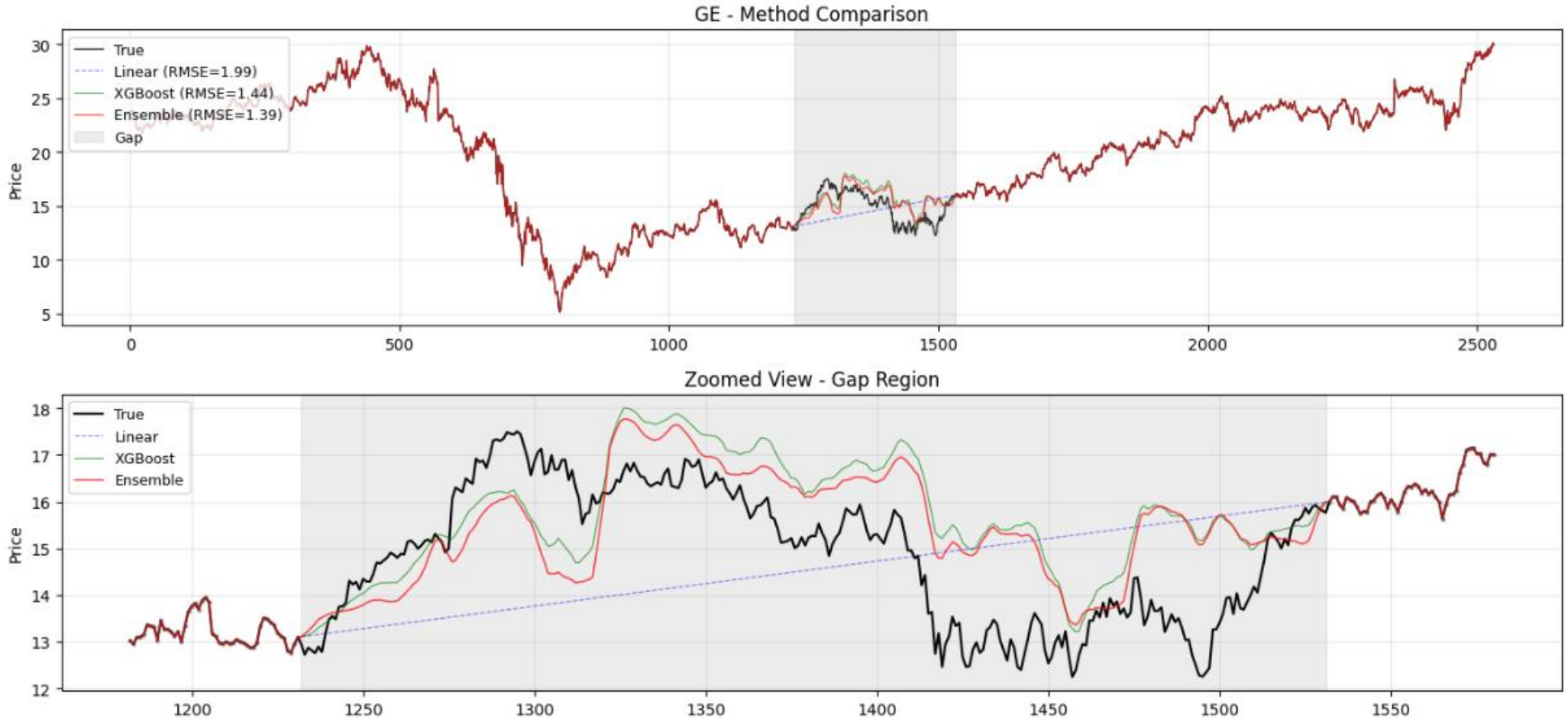


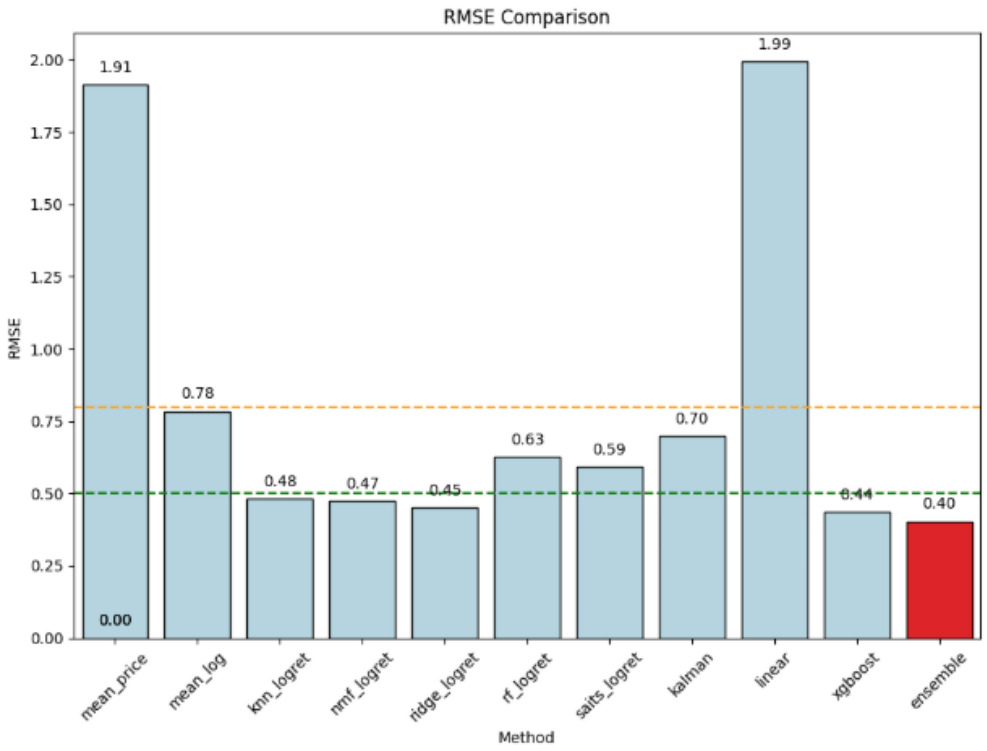



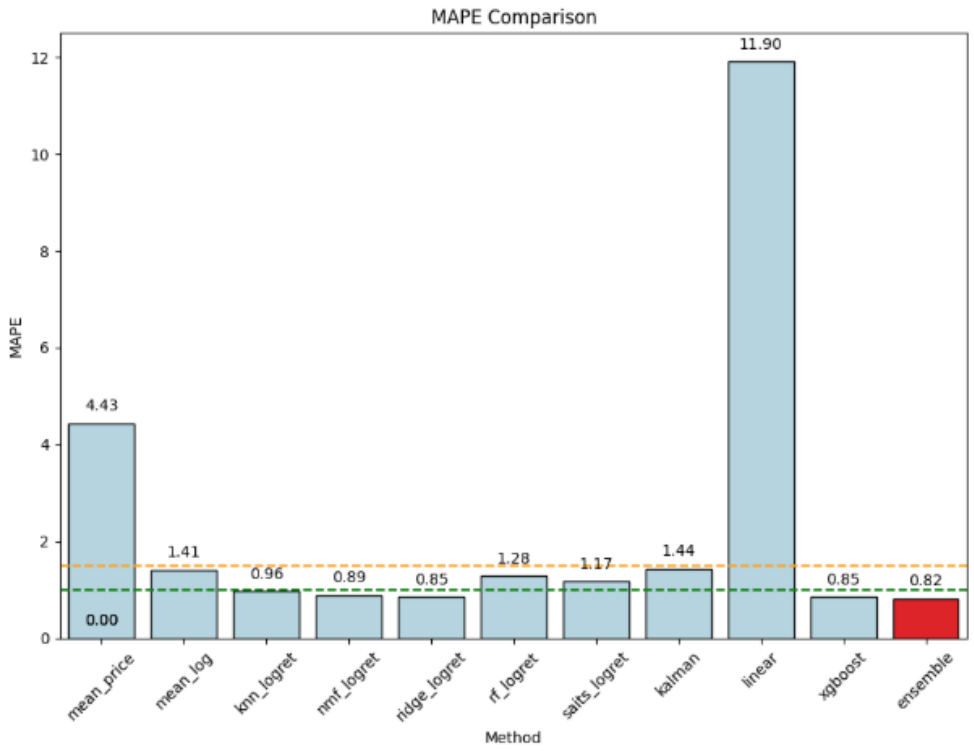
# Model – XGBOOST, Ensemble, Linear




- XGBOOST - Uses an ensemble of decision trees to learn the relationship between peer companies and GE
- Ensemble -Combines predictions from multiple models (XGBoost  Ridge Regression + GBM)

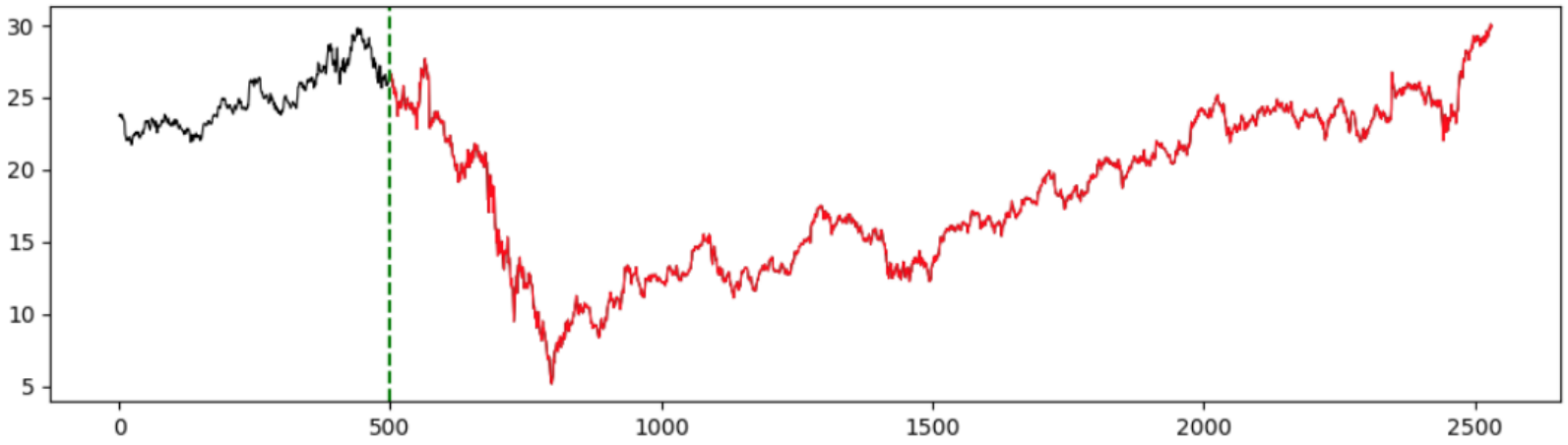
# Model – Performance Comparison

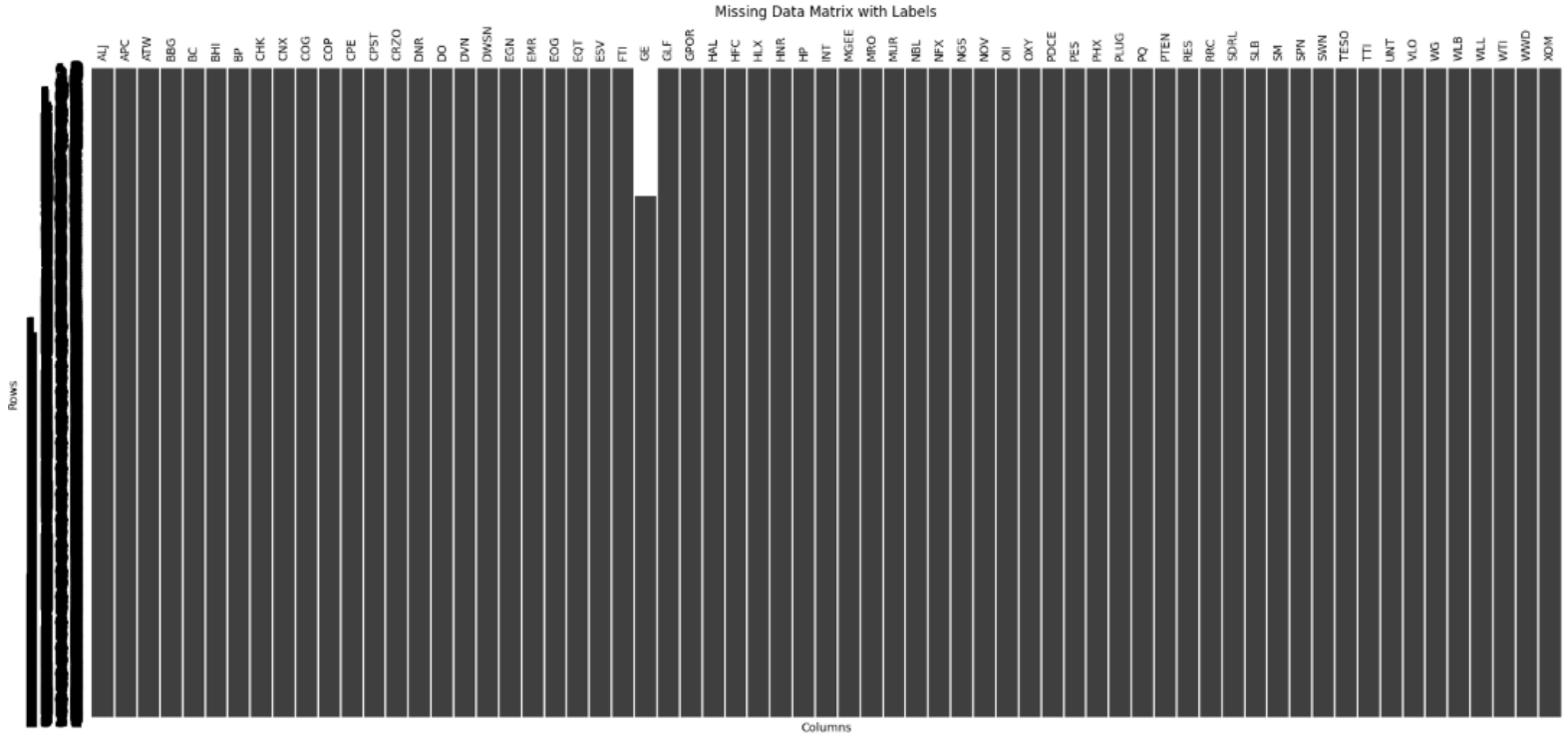


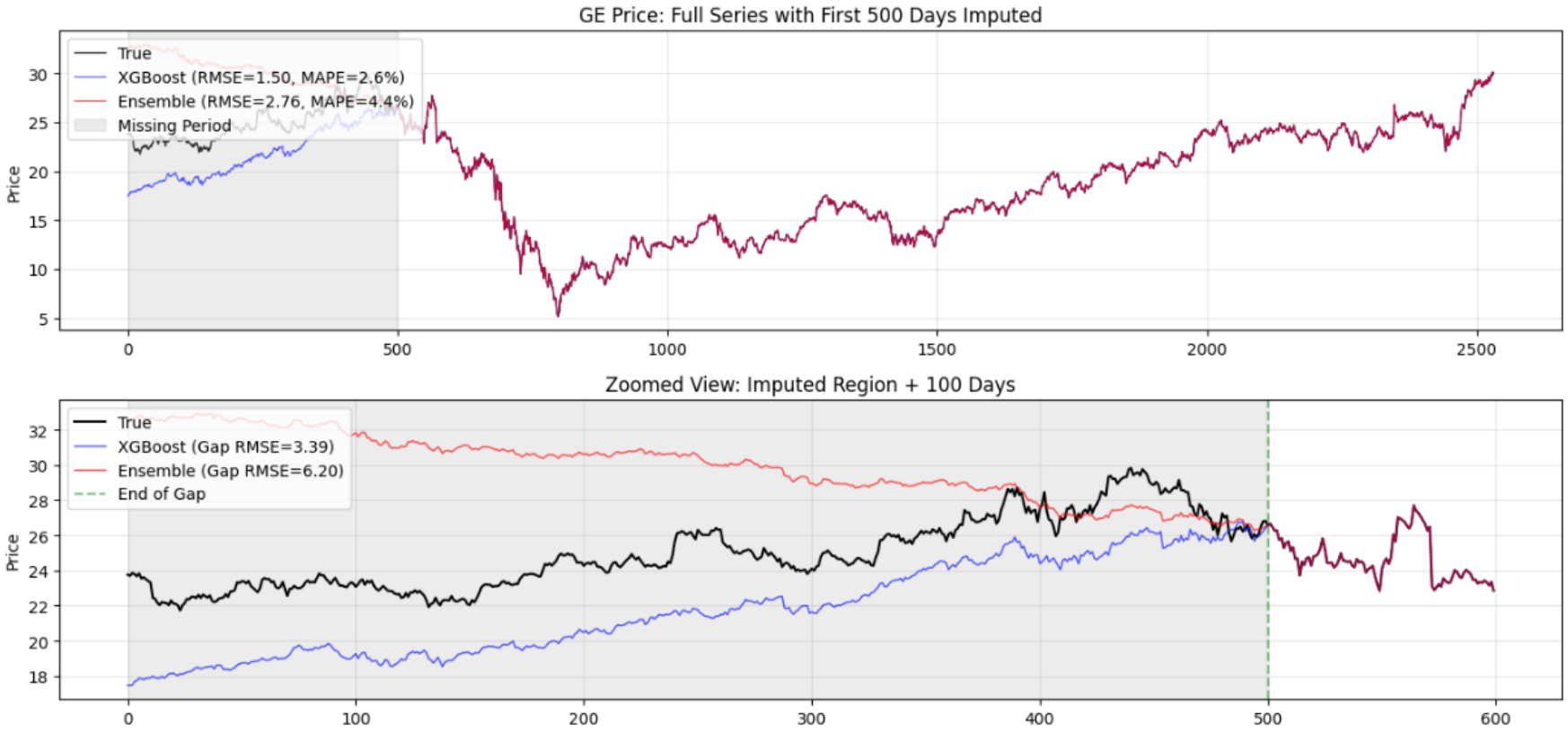

- **Log returns** consistently **outperform price-based methods** - **Working in log-return space is crucial for time series imputation**
- Ensemble methods and gradient boosting **- XGBoost and ensemble approaches leverage peer relationships effectively**
- **Simple can be powerful** - **Ridge regression** on log returns performs remarkably well despite its simplicity
- **Direct price imputation fails** - Mean prices and linear interpolation in price space perform terribly
- **Peer-based methods dominate** - Methods that leverage correlations (ensemble, XGBoost, ridge) outperform univariate methods

# Original Data – Artificially Created Data Gap := First 500 days



# Model – XGBOOST, Ensemble

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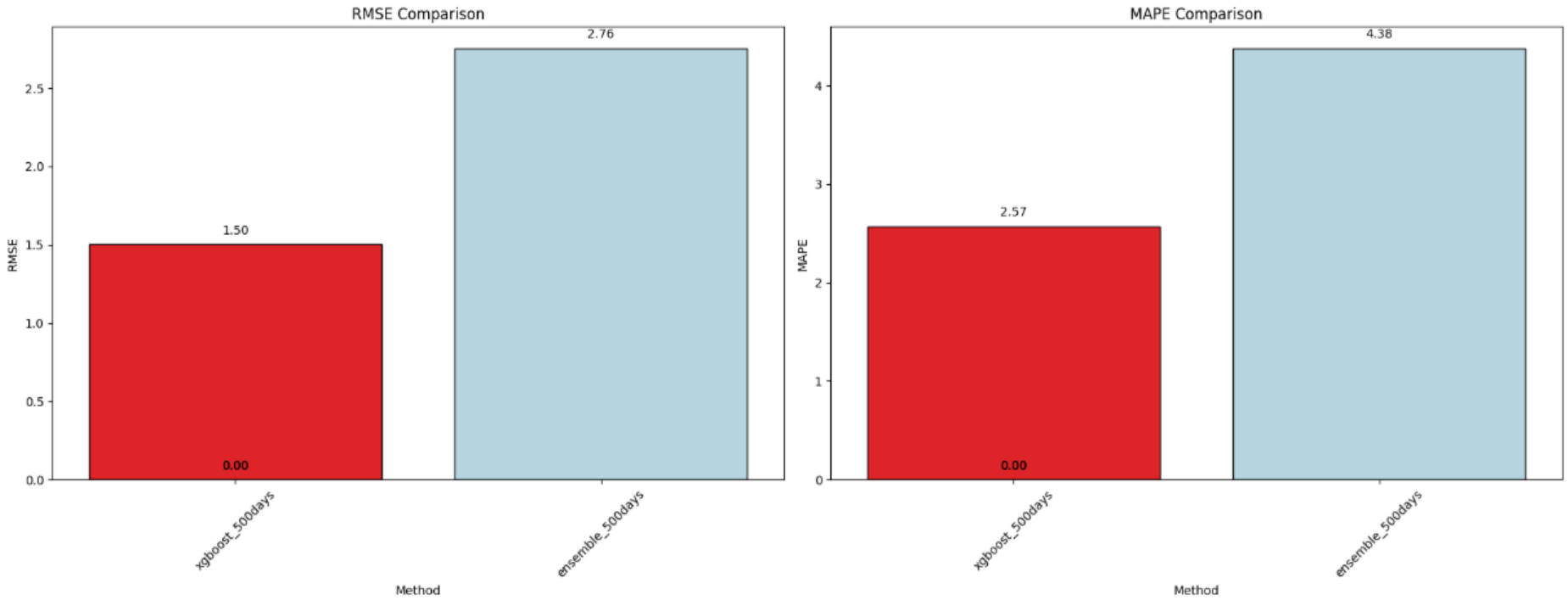


**XGBoost outperforms Ensemble significantly** - Unlike the mid-series gap where ensemble was best, XGBoost handles the start-of-series imputation better with ~46% lower RMSE

# MCAR Missingness – Some Models, Non-Exhaustive Summary

| Data | Method | Class |
|---|---|---|
| Price | Price fillna | Univariate |
| Log returns | Return fillna + inversion | Univariate |
| Log returns | Linear interpolation | Univariate |
| Log returns | Kalman smoothing (returns) | Univariate |
| Log returns | KNN | Multivariate |
| Log returns | MICE (ycimpute) | Multivariate |
| Log returns | miceforest (RF-MICE) | Multivariate |
| Log returns | XGBoost regression | Multivariate |
| Log returns | NMF | Multivariate |
| Log returns | SAITS / MIDA | Multivariate |